Question	Answer	Marks
6(a)(i)	W deflects upwards	B1
	W has deflection that is the mirror image of X	B1
	Y deflects downwards less than X **or** path of Y is parallel to particle Z	B1
6(a)(ii)	nucleus has most of the mass of the atom **or** nucleus is charged	B1
6(a)(iii)	atom is mostly empty space **or** nucleus occupies a <u>very</u> small proportion of the space in the atom	B1
6(b)	mass = 4 u	A1
	charge = (+)2e	A1
6(c)(i)	baryon	B1
6(c)(ii)	charge on up quark = $(+)\frac{2}{3}(e)$ **and** charge on down quark = $-\frac{1}{3}(e)$	C1
	(proton is up up down and neutron is up down down) $\left(2\times\frac{2}{3}\right)e-\left(1\times\frac{1}{3}\right)=1e$ **and** $\left(1\times\frac{2}{3}\right)e-\left(2\times\frac{1}{3}\right)e=0$	A1